

CSE 4345: Big Data Analytics

Week 7, Part 2 — Seminal Research & Case Study

Department of Computer Science & Engineering
Premier University, Chittagong

Semester 2026

Today's Agenda

- 1 Paper 1 — Thusoo et al. (2009): Hive at VLDB
- 2 Paper 2 — Thusoo et al. (2010): Hive at Petabyte Scale
- 3 Case Study — Spotify Discover Weekly (2015–2018)
- 4 Live Exercise

Seminal Papers Covered

- 1 Thusoo, A., et al. (2009). *Hive: A Warehousing Solution over a Map-Reduce Framework*. VLDB 2009.
- 2 Thusoo, A., et al. (2010). *Hive: A Petabyte Scale Data Warehouse Using Hadoop*. IEEE ICDE 2010.

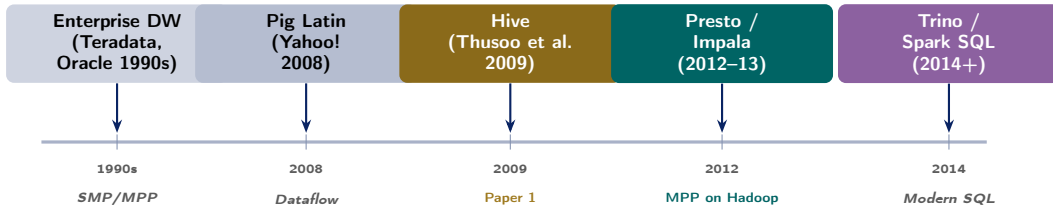
Case Study

Spotify Discover Weekly (2015–2018): HiveQL + Luigi pipelines over **2 billion playlists** to generate personalised weekly mixes for **30 million** listeners.

CLO Addressed

Paper 1 — Thusoo et al. (2009): Hive at VLDB

Research Landscape: SQL-on-Hadoop Timeline



Why Hive Was Necessary

By 2008 Facebook had **hundreds of analysts** who could write SQL but not Java MapReduce. Pig Latin was more expressive than MR but still procedural. Hive translated the *declarative* SQL dialect (HiveQL) directly to MapReduce — giving every analyst self-service access to Facebook's 15 TB/day event corpus without engineering involvement.

Thusoo et al. (2009): Publication Context

Full Citation

Thusoo, A., Sarma, J. S., Jain, N., Shao, Z., Chakka, P., Anthony, S., Liu, H., Wyckoff, P., & Murthy, R. (2009). **Hive: A Warehousing Solution over a Map-Reduce Framework**. *Proceedings of the VLDB Endowment*, 2(2), 1626–1629.

Venue: VLDB 2009, Lyon, France — Rank A* (CORE)

Cited by: >3,000 (Google Scholar, 2024)

Authors: Ashish Thusoo & Joydeep Sen Sarma (Facebook Data Infrastructure team)

The Founding Problem

Facebook's Hadoop cluster held **2 PB** of data by 2009 and ran **7,500 queries/day**. Analysts wrote ad hoc

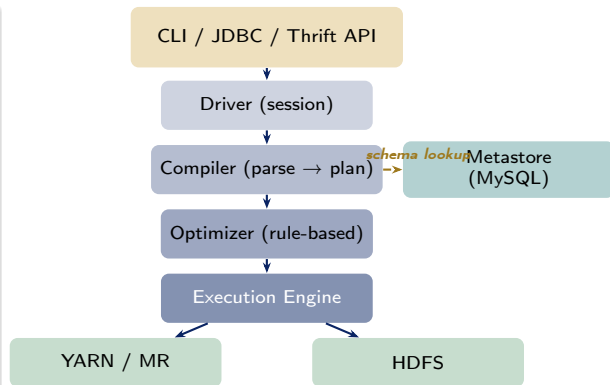
Hive's Design Goals (from the paper)

- 1 **Familiarity:** HiveQL must be close enough to ANSI SQL that experienced analysts need no retraining
- 2 **Extensibility:** custom SerDes and UDFs allow processing of any data format (logs, JSON, Thrift)
- 3 **Performance:** partition pruning and bucketing must reduce the data scanned for common query patterns
- 4 **Fault tolerance:** rely on MapReduce / HDFS for durability; Hive adds no new failure modes

Technical Contribution 1 — Hive Architecture

Four-Component Design

- **Metastore:** a relational DB (MySQL or Derby) storing table schemas, partition metadata, column statistics, and SerDe mappings. Shared across all Hive clients — the “schema-on-read” catalogue for HDFS files.
- **Driver:** receives HiveQL queries, manages the session, maintains query lifecycle
- **Compiler:** parses HiveQL → abstract syntax tree → semantic analysis (type checking, resolving column names against Metastore) → logical plan → physical plan (MapReduce DAG)
- **Execution Engine:** submits MR jobs to YARN in dependency order; collects results



Technical Contribution 2 — Metastore & SerDe

Metastore: Schema-on-Read Catalogue

The Hive Metastore stores:

- **Table definitions:** column names, types, location in HDFS, file format, compression codec
- **Partition metadata:** for a table partitioned on (*dt*, *country*), the Metastore holds the HDFS path for each (*dt*, *country*) value pair; enables partition pruning without listing all files
- **Bucket count:** number of buckets per partition; used to decide whether a sort-merge join avoids a shuffle
- **Column statistics:** NDV, min/max, nulls; used by cost-based optimizer (Hive 0.14+, 2015)

SerDe: Pluggable Format Layer

A **SerDe** (Serializer / Deserializer) translates raw bytes on HDFS into Hive rows and back.

SerDe	Format
LazySimpleSerDe	\t-delimited text
OpenCSVSerDe	RFC 4180 CSV
JsonSerDe	JSON (one object/line)
AvroSerDe	Avro binary + schema
ParquetHiveSerDe	Columnar Parquet
OrcSerDe	Columnar ORC
RegexSerDe	Any log via regex

UDFs: Arbitrary Extension

Paper 2 — Thusoo et al. (2010): Hive at Petabyte Scale

Thusoo et al. (2010): ICDE and Production Results

Full Citation

Thusoo, A., Sarma, J. S., Jain, N., Shao, Z., Chakka, P., Zhang, N., Antony, S., Liu, H., & Murthy, R. (2010). **Hive: A Petabyte Scale Data Warehouse Using Hadoop**. In *Proceedings of the 26th IEEE International Conference on Data Engineering (ICDE)*, pp. 996–1005. Long Beach, CA.

Venue: IEEE ICDE 2010 — Rank A (CORE)

Cited by: >1,500 (Google Scholar, 2024)

Extends: Thusoo et al. 2009 with Facebook's production Hive deployment at petabyte scale

Facebook Production Scale (2010)

Metric	Value
Data stored	2 PB
Daily growth	15 TB
Daily queries	7,500+
Active Hive users	200+ analysts
Longest query	48 hours
Most used feature	Partitioning

New Contributions Over the 2009 Paper

- Detailed **query optimisation** techniques: map-side aggregation, skew join handling, and

Key Optimization: Map-Side Aggregation

For `SELECT country, COUNT(*) FROM`

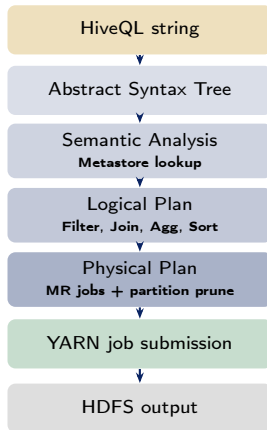
HiveQL Query Lifecycle — From SQL to MapReduce

Example: Partitioned Join Query

```
SELECT u.country, COUNT(*) AS plays FROM play_events  
p JOIN users u ON p.user_id = u.user_id WHERE p.dt =  
'2024-10-01' AND p.country IN ('BD','IN','US') GROUP BY  
u.country ORDER BY plays DESC;
```

Compiler Steps

- 1 **Parse:** HiveQL → AST
- 2 **Semantic Analysis:** resolve `p.user_id` to `play_events.user_id` : BIGINT; check `dt` is a partition column
- 3 **Logical Plan:** Filter → HashJoin → GroupBy → OrderBy
- 4 **Partition Pruning:** only load
`dt=2024-10-01/country=BD/, .../IN/,`
`/US/` (skip all other 365×200 partitions)



Hive's Lasting Contributions

- **Hive Metastore** became the de facto standard catalogue for all Hadoop-ecosystem tools: Spark, Presto/Trino, Impala all read from it
- **ORC format** (2013): Hive team developed Optimized Row Columnar; 2–4× faster reads than SequenceFile; predicate pushdown within stripes
- **LLAP** (Live Long and Process, 2016): Hive-on-Tez with persistent in-memory daemon; sub-second interactive queries on multi-TB tables
- **Hive-on-Spark** (2015): execution engine swapped from MR to Spark; same HiveQL, faster execution

Stanford CS246 / CMU 15-721 Perspective

- CMU 15-721 presents Hive as the first **declarative interface over a distributed execution engine** — separating the *what* (HiveQL) from the *how* (MR DAG). The same separation reappears in Spark SQL's Catalyst and Presto's connector model.
- Stanford CS246 uses Hive as a benchmark baseline for **join optimisation**: when to broadcast a small table (map-side join) vs. sort-merge join vs. hash join — a question that drives Presto's cost-based optimizer decisions

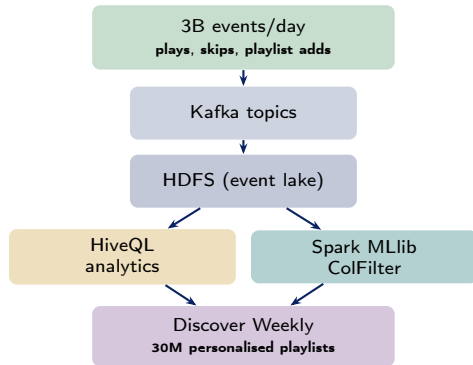
Case Study — Spotify Discover Weekly (2015–2018)

Spotify's Data Challenge

Scale of the Music Listening Graph

Spotify (founded 2006, 75M users by 2015) generates:

- **Streaming events:** every song play, skip, pause, repeat, and playlist add logged as a Kafka event — ≈ 3 billion events per day
- **30 million songs** in the catalogue; each with audio features (tempo, key, danceability, acousticness) computed by audio ML models
- **2 billion user-created playlists:** implicit feedback on what songs belong together
- All event data landed on HDFS via Kafka consumers and queried daily via HiveQL for product analytics, A/B testing, and recommendation training



Discover Weekly: The Recommendation Pipeline

Three-Signal Collaborative Filtering

Discover Weekly (launched July 2015) combines three data sources, all prepared via HiveQL on Hive + Spark:

- 1 **Playlist co-occurrence** (implicit feedback):
If songs A and B appear together in $>N$ playlists, they are “similar.” **HiveQL query:** `SELECT song_a, song_b, COUNT(*) FROM playlist_songs GROUP BY song_a, song_b HAVING COUNT(*) > 5`
Over **2 billion playlists** on Hive.
- 2 **NLP on artist metadata** (Natural Language Processing on music blog text) — **Word2Vec** on song co-occurrence sequences, like word2vec on sentences.
- 3 **Audio feature similarity** — CNN-derived

Scale Numbers (2015–2016)

Metric	Value
Weekly listeners served	30 million
Playlists analysed	2 billion
Songs in catalogue	30 million
Song-pair co-occurrences	>1 trillion
HiveQL query for co-occur.	~ 2 hours
Spark ALS matrix size	$75M \times 30M$
Weekly pipeline run time	~ 24 hours
User retention uplift	+40% playlist saves

Luigi DAG

Lessons Learned & Discussion Questions

Key Lessons from Spotify's Case

- 1 **Implicit feedback at scale beats ratings:**
Spotify's 2 billion playlists were richer signal than explicit star ratings (cf. Netflix Prize); collecting behavioural events via Kafka + HDFS is the enabling infrastructure
- 2 **HiveQL for feature engineering, Spark for ML:** co-occurrence aggregation (SQL) and ALS matrix factorisation (iterative) suit different engines; a single pipeline tool (Luigi) bridges them
- 3 **Schema design determines query speed:**
partitioning `play_events` by week reduced the co-occurrence HiveQL from 8 hours to 2 hours
- 4 **Cold-start is still hard:** new songs with no playlist appearances get audio-features

Discussion Questions (CLO3, CLO4)

- 1 Spotify's co-occurrence query runs `GROUP BY song_a, song_b` over 2 billion playlist rows. Estimate the output size if there are 30M songs. Propose a **partitioning + bucketing** scheme for the `playlist_songs` Hive table that would reduce the shuffle size for this query. [CLO4 — Apply]
- 2 The ALS collaborative filtering model is a matrix factorisation over a 75M-user $\times$ 30M-song matrix. At 4 bytes per float and rank $k = 50$, calculate the model size and explain whether it fits in a 5,000-node YARN cluster's aggregate RAM. [CLO4 — Analyse]

Live Exercise

Live Exercise — HiveQL on Local Hadoop (or SparkSQL)

Task (25 minutes, pairs)

Simulate a Hive session using **PySpark's SQL interface** (which reads the same Hive Metastore).

Dataset: a 10,000-row synthetic `play_events` CSV (generate with the skeleton below).

Goals:

- 1 Create a partitioned Spark table (partition on `dt`).
- 2 Run the co-occurrence query and time it.
- 3 Add a **broadcast join** hint for the songs dimension table and compare timings.
- 4 Use `EXPLAIN` to print the physical plan and identify where partition pruning occurs.

PySpark SQL Skeleton

```
from pyspark.sql import SparkSession
import random, datetime

spark = SparkSession.builder \
    .appName("HiveDemo") \
    .getOrCreate()
spark.setLogLevel("WARN")

# Generate synthetic data
rows = []
for i in range(10_000):
    rows.append({
        "user_id":    random.randint(1, 500),
        "song_id":    random.randint(1, 200),
        "playlist_id": random.randint(1, 1000),
        "dt":         "2024-10-01"
    })

df = spark.createDataFrame(rows)
df.createOrReplaceTempView("play_events")

# Co-occurrence query
result = spark.sql("""
```

References

Seminal Papers

- Thusoo, A., et al. (2009). Hive: A warehousing solution over a map-reduce framework. *Proc. VLDB Endowment*, 2(2), 1626–1629.
- Thusoo, A., et al. (2010). Hive: A petabyte scale data warehouse using Hadoop. *Proc. IEEE ICDE*, 996–1005.

Textbooks [TW], [SA], [TE]

- **[TW]** White, T. (2015). *Hadoop: The Definitive Guide*, 4th ed. (Ch. 17 — Hive). O'Reilly.
- **[SA]** Acharya, S., & Chellappan, S. (2015). *Big Data and Analytics* (Ch. 7 — Data Warehousing on Hadoop). Wiley.
- **[TE]** Erl, T., Khattak, W., & Buhler, P. (2016). *Big Data Fundamentals* (Ch. 9 — SQL-on-Hadoop). Prentice Hall.

Case Study Sources

- Spotify Engineering. (2015). *How Spotify Discover Weekly Works*. labs.spotify.com/2015/08/06/discover-weekly-works/

Key Takeaways

- Thusoo et al. (2009) gave the world a **declarative interface over distributed storage** — HiveQL's impact was democratisation: 200 analysts replaced 20 Java engineers
- The **Metastore** is Hive's most durable contribution — it outlived MR-based Hive execution and is now the universal catalogue for Spark, Presto, Trino, Athena, and Delta Lake
- Spotify's case demonstrates that **SQL (HiveQL) and iterative ML (Spark ALS)** are complementary in a recommendation pipeline — neither alone is sufficient at billion-playlist scale
- **Schema design is a first-class skill**: the right partition column (week vs. song vs. user) changes query latency by hours on petabyte data

Quote of the Week

"Hive provides a simple and optimizable query language, familiar to all data warehouse users, while retaining the scalability of Hadoop."

— Thusoo et al. (2009)

Part 1 Preview — Week 8

Kafka, Sqoop, Flume, Lambda/Kappa Architecture

- Kafka topics, partitions, consumer groups
- Exactly-once semantics
- Lambda: batch + speed + serving layers